# *Flow of Groundwater in Relatively Thick Leaky Aquifers*

MAHDI S. HANTUSH

*New Mexico Institute of Mining and Technology, Socorro*

*Abstract.* Solutions to many problems of flow in leaky aquifers are based on a differential equation that includes a leakage term in addition to the usual terms of the equation of groundwater motion. In other words, the effect of leakage on the flow is incorporated in the differential equation rather than expressed as a boundary condition, as it actually occurs in the physical system. Such an approximation is justified when the main aquifer is relatively thin, provided, of course, that the conductivity of the semipervious layer relative to those of the main aquifers of the leaky system is small. Although the latter condition is almost always prevalent in nature, the aquifers are not always relatively thin. In such aquifers, the use of the available solutions may yield poor results, especially when a quantitative criterion for the relative thickness of the aquifers is not available. Solutions are developed by using the usual differential equation of groundwater motion and the pertinent initial and boundary conditions, expressing the effect of leakage on the flow as a boundary condition. Moreover, a quantitative criterion is established for the applicability of the solutions that are already available from the approximate theory now in use. The already available solutions appear to be applicable wherever $b/B < 0.10$, where $b$ is the thickness of the main aquifer, and $B = \sqrt{T/(K'/b')}$ is the leakage factor; $T$ is the transmissivity of the aquifer; and $(K'/b')$ is the leakage coefficient of the flow system. (Key words: Groundwater; hydrologic systems; leaky aquifers; seepage; wells)

## INTRODUCTION

Aquifers that gain or lose water through over- and/or underlying semipervious layers are termed leaky aquifers. The flow in such aquifers is three-dimensional in nature. A general mathematical treatment for the flow in such systems is complicated. Mathematical analyses under certain conditions [see for example, *Jacob*, 1946; *Hantush and Jacob*, 1955; *Hantush*, 1957, 1960, 1964, 1967] have, however, resulted in many simple and relatively good approximations that describe this flow phenomenon. One of the main assumptions is that the hydraulic conductivities of the semipervious layers are so small relative to those of the main aquifers [a condition that is of common occurrence in nature] that the flow is essentially vertical in the semipervious layers and almost horizontal in the main aquifers. As a consequence of this assumption, the rate of leakage gained or lost by a main aquifer at its interface with the semipervious layer has been assumed to be uniformly distributed along the thickness of the main aquifer if this thickness is relatively thin. In other words, the actual flow system is replaced by a hypothetical system of completely impermeable confining beds, in which the flow is augmented or diminished at rate per unit volume of the main aquifer equal to the rate of leakage per unit thickness of this aquifer. Accordingly, a differential equation incorporating the effect of leakage has been derived to approximate the flow in such leaky systems [see for example, *Jacob*, 1946; *Hantush*, 1957, 1964, 1967]. Such a model may yield poor results when the main aquifers of the leaky system are not relatively thin. In such cases, the mathematical analysis should be based on the usual differential equation of groundwater motion, expressing the leakage across the interface as a boundary condition. In other words, the analysis should be based, in addition to other pertinent boundary conditions, on

$$\nabla^2 h_n = 1/\nu_n \, \partial h_n/\partial t$$

and

$$K_n \, \partial h_n/\partial z = (K'/b')(h_n - h_{n+1})$$

where the latter expression is evaluated at the interface between the semipervious layer and the $n$th aquifer; $\Delta^2$ is the Laplacian operator; $h_n$, $h_{n+1}$ are, respectively, the heads in the $n$th and $(n+1)$th aquifer; $K_n$ is the conductivity of the $n$th aquifer; $K'$, $b'$ are, respectively, the conductivity and the thickness of the semipervious layer; and $\nu_n = K_n/S_{ns}$, $S_{ns}$ is the specific storage of the $n$th aquifer.

The subsequent treatment is an example of such an analysis. In addition to yielding more general flow formulas, the analysis provides a criterion for the applicability of the solutions that have been already obtained through use of the approximate mathematical model.

## FLOW TO WELLS IN THICK LEAKY AQUIFERS

Figure 1 is a diagrammatic representation of wells penetrating a leaky system. The aquifers of the system are individually elastic, homogeneous, isotropic, and uniform in thickness, and the tangent of the angle of tilt is less than 1%. The conductivity, the storativity, and the leakage coefficient remain constant with time and in the space of the layer they characterize. The storage in the semipervious layer is negligible; the conductivity of the semipervious layer is so small relative to that of the main aquifers (less than 0.01) that the flow in this layer is essentially vertical, and the head in the aquifer supplying leakage remains uniform. Consequently, the rate of leakage crossing the interface between the pumped aquifer and the semipervious layer is directly proportional to the drawdown at this interface. The system is of an infinite areal extent. When the effect of the pumping does not reach the outer boundaries of a system, the system is effectively infinite in areal extent.

The piezometric drawdown induced around a well that is partially penetrating and steadily discharging from the main aquifer of the system of Figure 1, when the well is perforated throughout its depth of penetration, can be described (see Figure 1 for coordinate system) by the following boundary-value problem:

$$\partial^2 s/\partial r^2 + 1/r\ \partial s/\partial r + \partial^2 s/\partial z^2 = 1/\nu\ \partial s/\partial t \quad (1)$$

$$s(r, z, 0) = 0 \qquad s(\infty, z, t) = 0 \quad (2)$$

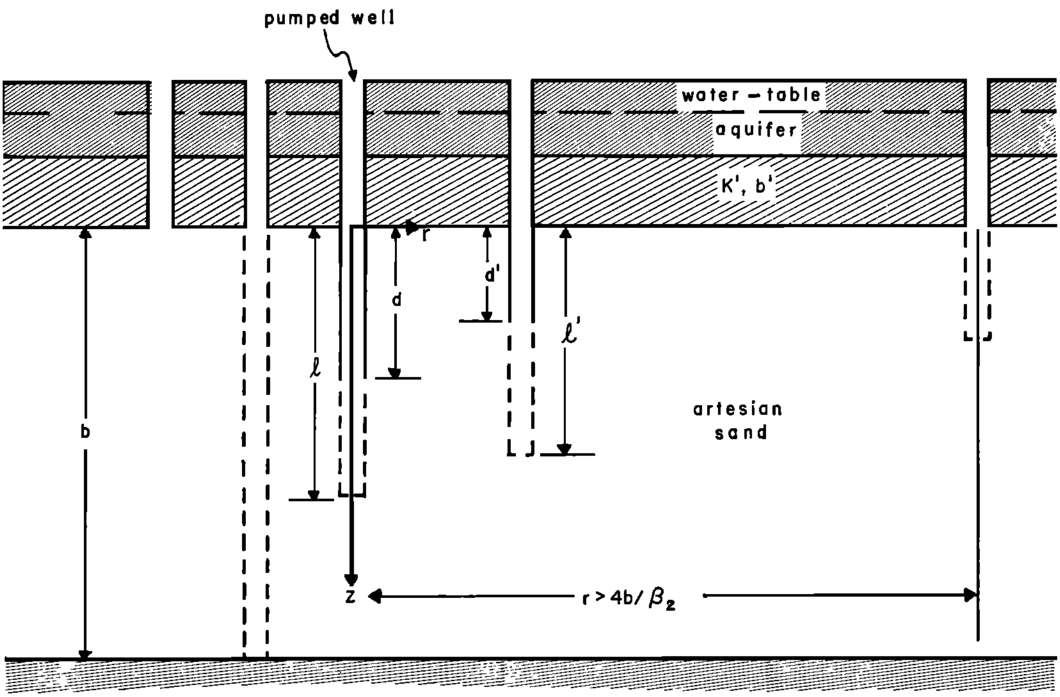


Fig. 1. Diagrammatic representation of wells partially penetrating a leaky aquifer.

$$L_{r\to 0} r\, \partial s/\partial r = -Q/2\pi K l \quad 0 < z < l \qquad (3)$$
$$= 0 \qquad l < z < b$$

$$(K'/b')s(r, 0, t) = K\, \partial s/\partial z(r, 0, t) \qquad (4)$$

$$\partial s/\partial z(r, b, t) = 0 \qquad (5)$$

in which

$$\nu = K/S_s = T/S \qquad (6)$$

$b$ = the thickness of the pumped aquifer, $L$;
$b'$ = the thickness of the semipervious layer, $L$;
$K$ = the hydraulic conductivity of the pumped aquifer, $L/T$;
$K'$ = the hydraulic conductivity of the semipervious layer, $L/T$;
$K'/b'$ = the leakage coefficient, $1/T$;
$l$ = the depth of penetration of the pumped well, $L$;
$Q$ = the constant discharge of the well, $L^3/T$;
$r$ = radial distance in horizontal planes measured from the center of the well [see Figure 1 for coordinate systems], $L$;
$S$ = $bS_s$, the storativity (storage coefficient) of the aquifer;
$S_s$ = the specific storage of the aquifer, $1/L$;
$s$ = the induced piezometric drawdown, $L$;
$T$ = $Kb$ = the transmissivity of the aquifer, $L^2/T$;
$t$ = time since pumping began, $T$;
$z$ = the vertical coordinate [see Figure 1 for coordinate systems], $L$.

Applying the Laplace transformation with respect to $t$ and the zero-ordered Hankel transformation with respect to $r$ [see, for example, *Carslaw and Jaeger,* 1959; *Hantush,* 1960] on the preceding boundary-value problem, then solving the resulting boundary-value problem will ultimately yield

$$\bar{V}(\alpha, z, p) = c_1\left(a\sqrt{p/\nu + \alpha^2} \cosh z\sqrt{p/\nu + \alpha^2} + \sinh z\sqrt{p/\nu + \alpha^2}\right) + Q/2\pi K l p\left(1 - \cosh z\sqrt{p/\nu + \alpha^2}\right) \quad \text{for} \quad 0 \le z \le l \qquad (7)$$

and

$$\bar{V}(\alpha, z, p) = c_2 \cosh (z - b)\sqrt{p/\nu + \alpha^2} \quad \text{for} \quad l \le z \le b \qquad (8)$$

where

$$a = K/(K'/b') \qquad (9)$$

$\bar{V}(\alpha, z, p)$ denotes the zero-ordered Hankel transform with respect to $r$ of the Laplace transform with respect to $t$ of the function $s(r, z, t)$, and $\alpha$ and $p$ are the parameters of these transformations, respectively.

At $z = l$, the values of $\bar{V}$ and $K\partial V/\partial z$ as obtained from (7) are, respectively, equal to those obtained from (8). The use of these conditions determines the values of $c_1$ and $c_2$. After obtaining these values, (7) and (8) become, respectively

$$\bar{V} = \frac{(Q/2\pi K l)}{p[p/\nu + \alpha^2]}\left\{\frac{[\sinh b\surd - \sinh (b - l)\surd][\sinh z\surd + a\surd \cosh b\surd]}{[\cosh b\surd + a\surd \sinh b\surd]} + (1 - \cosh z\surd)\right\} \quad \text{if} \quad 0 \le z \le l \qquad (10)$$

and

$$\bar{V} = \frac{Q}{2\pi K l}\, \frac{(\cosh l\surd + a\surd \sinh l\surd - 1) \cosh (b - z)\surd}{p[p/\nu + \alpha^2][\cosh b\surd + a\surd \sinh b\surd]} \quad \text{if} \quad l < z < b \qquad (11)$$

in which

$$\surd = \sqrt{p/\nu + \alpha^2}$$

Obtaining the inverse Laplace transform (using the translation, the convolution, and the residue theorems) of (10) and (11), the inverse zero-ordered Hankel transform of the resulting expressions will then ultimately yield [noting that the inversion of (10) happens to be equivalent to the inversion of (11)] the equation for the piezometric drawdown. In the process of this evaluation, the following relations have been used:

$$\int_0^\infty e^{-\nu\alpha^2\tau}\alpha J_0(\alpha r)\,d\alpha = (1/2\nu\tau)\exp(-r^2/4\nu\tau)$$

$$\int_0^t \exp(-r^2/4\nu\tau - \nu\tau\beta^2/b^2)\,d\tau/\tau = W(r^2/4\nu\tau\; r\beta/b)$$

and

$$\beta\tan\beta = b/a$$

where $J_0$ is the zero-order Bessel function of the first kind, and $W(u, x)$ is the well function for leaky aquifers.

## DRAWDOWN AROUND PARTIALLY PENETRATING WELLS

*Piezometric drawdown.* The results of the preceding evaluation gives the piezometric drawdown around a steadily discharging well that is *perforated throughout its depth of penetration.* It can be finally expressed as

$$s(r, z, t) = \frac{Q}{\pi K l}\left\{R_1 \cos\frac{b-z}{b}\beta_1 W\left(u, \frac{r}{b}\beta_1\right) + \sum_{m=2}^{\infty} R_m \cos\frac{b-z}{b}\beta_m W\left(u, \frac{r}{b}\beta_m\right)\right\} \quad (12)$$

in which

$$u = r^2/4\nu t \quad (13)$$

$$R_m = \frac{\sin\beta_m - \sin(b-l)\beta_m/b}{2\beta_m + \sin 2\beta_m} \quad (14)$$

$R_1$ is given by (14), with $m = 1$; $W(u, x)$ is the well function for leaky aquifers, tabular values for which are available [*Hantush*, 1956, 1964; *DeWiest*, 1965; *Schoeller*, 1962], and $\beta_m$ are the zeroes of

$$\beta_m \tan\beta_m = (b/a) = (b/B)^2 \quad (15)$$

where $B$ denotes the leakage factor and is given by $B^2 = T/(K'/b')$.

The first six zeroes of (15) are given by *Carslaw and Jaeger* [1959, p. 491]. The remaining zeroes can be approximated by

$$\beta_m = \beta_6 + (m-1)\pi \qquad m = 7, 8, 9, \cdots \quad (16)$$

For $(b/B)^2 \leq 0.2$, the roots of (15) can be approximated by

$$\beta_1 = (1 - c/6)\sqrt{c} \quad (16a)$$

and

$$\beta_m = (m-1)\pi + \frac{c}{(m-1)^2\pi^2} - \frac{c^2}{(m-1)^3\pi^3}, \qquad m = 2, 3, 4 \quad (16b)$$

where

$$c = (b/B)^2$$

*Piezometric drawdown for relatively large time ($t > 5bS/K\beta_2^2$).* For $u \leq x^2/20$, the function $W(u, x)$ can, for all practical purposes, be replaced by $2K_0(x)$. Thus, for $u < (r\beta_m/b)^2/20$, that is, for $t \geq 5bS/K\beta_2^2$, where $\beta_2$ is the second zero of (15), equation 12 can be approximated by

$$s = \frac{Q}{\pi K l}\left\{R_1 \cos\frac{b-z}{b}\beta_1 W\left(u, \frac{r}{b}\beta_1\right) + 2\sum_{m=2}^{\infty} R_m \cos\frac{b-z}{b}\beta_m K_0\left(\frac{r}{b}\beta_m\right)\right\} \quad (17)$$

in which $K_0$ is the zero-order modified Bessel function of the second kind, tabular values for which are available [*Dwight*, 1961].

*Steady-state piezometric drawdown.* This drawdown is given by (17) as $t \to \infty$. As $t \to \infty$, $u \to 0$ and $W(0, r\beta_1/b) = 2K_0(r\beta_1/b)$. Consequently, the steady-state drawdown is given by

$$s = Q/\pi K l\;\{\text{the braced term of (17), with } 2K_0(r\beta_1/b) \text{ replacing } W\} \quad (18)$$

*Piezometric drawdown at $r > 4b/\beta_2$.* For $x > 4$, the function $W(u, x)$ and $K_0(x)$ approach zero. Consequently, the series of (12) and (17) will be of insignificant numerical value relative to that of $W(u, r\beta_1/b)$. Thus, for $(r\beta_2/b) > 4$, the series in (12) and (17) may be safely neglected. Consequently, (12) and (17) will both become

$$s = \frac{Q}{\pi K l}\left\{R_1 \cos\frac{b-z}{b}\beta_1 W\left(u, \frac{r}{b}\beta_1\right)\right\} \quad (19)$$

The corresponding *steady-state equation* is

$$s = \frac{Q}{\pi K l}\left\{2R_1 \cos\frac{b-z}{b}\beta_1 K_0\left(\frac{r}{b}\beta_1\right)\right\} \quad (20)$$

where $R_1$ is defined by (14), with $m = 1$.

*Piezometric drawdown for a well perforated between the depths d and l.* If the discharging well is perforated between the depths $d$ and $l$ (see Figure 1), the counterpart of any of the preceding drawdown equations (which are for a well perforated throughout its depth of penetration) can be shown to be given by

$$\mathfrak{s} = Q/\pi K(l - d)\{Z(l) - Z(d)\} \tag{21}$$

where $Z(l)$ is the braced term of the right-hand member of that equation, and $Z(d)$ is the value of $Z(l)$, with $d$ replacing $l$.

*Drawdown at the face of the well.* The drawdown at the face of a pumping well that is perforated throughout its depth of penetration can be approximated [*Hantush*, 1964] by (12) or (17), whichever applies, with $z = 0$ and $r = r_w$, $r_w$ being the effective radius of the well. For a well that is perforated between the depths $d$ and $l$, the drawdown at the face of the well can be approximated by (21), with $z = 0.5(l + d)$ and $r = r_w$.

*Average drawdown.* The average drawdown in an observation well that is perforated between the depths $d'$ and $l'$ can be obtained by integrating the equation of drawdown in piezometers [Equations 12, 17-20] with respect to $z$ between the limits $d'$ and $l'$, then dividing the results by $l' - d'$ [*Hantush*, 1964]. The process is straightforward, since it involves integration of cosine terms only. For example, the drawdown in an observation well that is *perforated throughout the thickness of the aquifer* (the limits of integration are 0 and $b$) will be obtained from (12) as

$$\bar{s} = \frac{Q}{\pi K l}\left\{R_1 \frac{\sin \beta_1}{\beta_1} W\left(u, \frac{r}{b}\beta_1\right) + \sum_{m=2}^{\infty} R_m \frac{\sin \beta_m}{\beta_m} W\left(u, \frac{r}{b}\beta_m\right)\right\} \tag{22}$$

where $\bar{s}$ denotes the average drawdown in the observation well.

## DRAWDOWN AROUND COMPLETELY PENETRATING WELLS

The drawdown distribution around a completely penetrating pumping well can be obtained from the equations of the preceding sections after replacing $l$ by $b$ and $d$ by zero. For example, the piezometric drawdown around a steady well that completely penetrates the aquifer, from (12) with $l = b$, is given by

$$s(r, z, t) = \frac{Q}{\pi K b}\left\{\frac{\sin \beta_1 \cos (b - z)\beta_1/b}{2\beta_1 + \sin 2\beta_1} W\left(u, \frac{r}{b}\beta_1\right) + \sum_{m=2}^{\infty} \frac{\sin \beta_m \cos (b - z)\beta_m/b}{2\beta_m + \sin 2\beta_m} W\left(u, \frac{r}{b}\beta_m\right)\right\} \tag{23}$$

and the corresponding drawdown at $r > 4b/\beta_2$, from (19) with $l = b$, is given by

$$s(r, z, t) = \frac{Q}{\pi K b}\frac{\sin \beta_1 \cos (b - z)\beta_1/b}{2\beta_1 + \sin 2\beta_1} W\left(u, \frac{r}{b}\beta_1\right) \tag{24}$$

The counterparts of the other equations can be similarly obtained.

## COMPARISON WITH THE APPROXIMATE AVAILABLE SOLUTIONS

The solutions presented in the preceding sections depend on values of $\beta_m$, among other things. The values of $\beta_m$ [the zeroes of (15)] depend on the ratio $b/B$. For $(b/B) \leq 0.10$ or $(b/B)^2 \leq 0.01$, the following approximations hold: $\beta_1 \simeq b/B$ and $\beta_m \simeq (m - 1)\pi$, $m = 2, 3, 4, \cdots$, $\sin \beta_1 \simeq \beta_1$, $\sin \beta_1(b - l)/b \simeq (b - l)/b$, $\sin 2\beta_1 \simeq 2\beta_1$, and $\cos (b - z)\beta_1/b \simeq 1$. Consequently, when $b/B \leq 0.10$ (a criterion that generally obtains in practice), the piezometric drawdown around a well that is perforated throughout its depth of penetration will, from (12) and the preceding approximations, reduce to

$$s = \frac{Q}{4\pi K b}\left\{W\left(u, \frac{r}{B}\right) + \frac{2b}{\pi l}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi l}{b} \cdot \cos\frac{n\pi z}{b} W\left(u, \frac{n\pi r}{b}\right)\right\} \tag{25}$$

which (on observing that, for $b/B < 0.10$, the quantity

$$\sqrt{(r/B)^2 + (n\pi r/b)^2} = (n\pi r/b)\sqrt{1 + (b/n\pi B)^2}$$

is very closely equal to $n\pi r/b$) is identical to the solution obtained by *Hantush* [1957], using the approximate equation of motion for leaky aquifers.

The drawdown around *completely penetrating wells,* if $b/B \leq 0.10$, will, from (23) and the abovementioned approximations, be given by

$$s = Q/(4\pi Kb)\; W(u, r/B) \qquad (25a)$$

which is the Hantush-Jacob formula as obtained by using the approximate theory of motion in leaky aquifers [*Hantush and Jacob,* 1955].

## SEEPAGE AND BANK STORAGE IN LEAKY AQUIFERS

Figure 2 is a schematic representation of leaky aquifers that are cut through by pervious stream channels. In Figure 2*a*, the leaky artesian aquifer is overlain by a semipervious layer on the top of which the head is maintained uniform. In Figure 2*b*, the leaky water-table aquifer overlies a semipervious layer at the bottom of which the head is maintained constant. In both cases the systems are semi-infinite in areal extent. If the water level in the channel is suddenly changed to a new permanent stage, the groundwater flow in these systems (assuming that the rise of the water-table relative to the initial depth of saturation for the water-table case is small) can be described (see Figures 2*a* and 2*b* for the *coordinate systems*) by the following boundary-value problem:

$$\partial^2 Z/\partial x^2 + \partial^2 Z/\partial z^2 = 1/\nu \quad \partial Z/\partial t \qquad (26)$$

$$Z(x, z, 0) = 0 \qquad (27)$$

$$Z(0, z, t) = \delta \qquad (28)$$

$$Z(\infty, z, t) = 0 \qquad (29)$$

$$K\, \partial Z/\partial z(x, 0, t) = (K'/b')Z(x, 0, t) \qquad (30)$$

$$\partial Z/\partial z(x, b, t) = 0 \qquad (31)$$

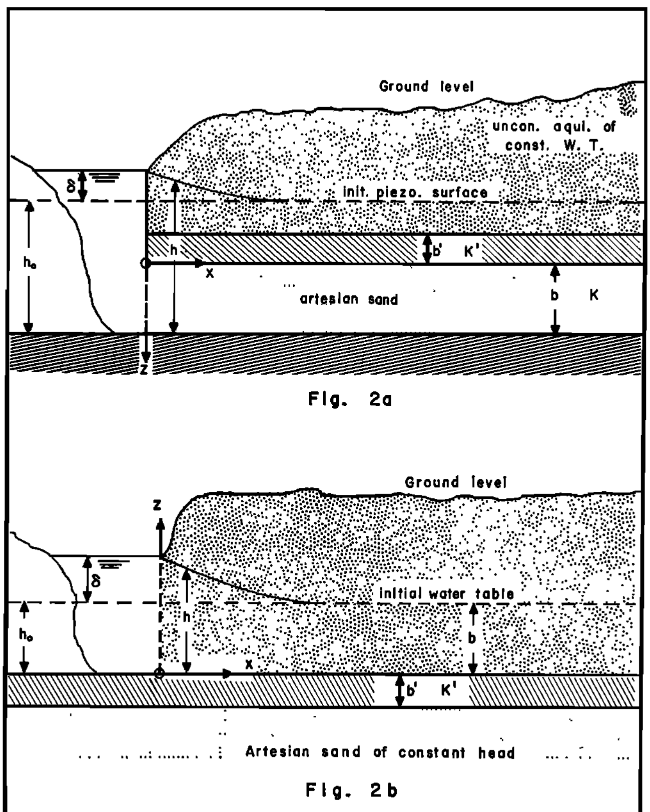


Fig. 2. Diagrammatic representation of streams cutting through leaky aquifers.

in which

$$Z = h - h_0$$

$$\nu = K/S_s \text{ (for the confined system)} \quad (32)$$

$$= K(2h_0 + \delta)/2\epsilon$$

$$\text{(for the unconfined system)} \quad (33)$$

where $h(x, z, t)$ is the piezometric head; $h_0$ is the water level that would have prevailed in the system if the sudden change of water level in the channel were not changed; $K$ is the conductivity of the aquifer; $S_s$ and $\epsilon$ are, respectively, the specific storage of the artesian aquifer and the specific yield of the water-table aquifer; and $\delta$ is the change in the stream water level.

The Laplace transformation with respect to $t$ and the Fourier sine transformation with respect to $x$ [*Carslaw and Jaeger,* 1959] are used in solving the preceding boundary-value problem. The process is straightforward. In the process of inverting the solution of the transformed problem, the convolution and the translation theorems in the Laplace transformation theory and the following integral relations have been used:

$$\frac{2}{\pi}\int_0^\infty \{1 - \exp[-\tau(\beta^2 + \omega^2)]\}\frac{\omega \sin \omega x}{\beta^2 + \omega^2}\, d\omega$$
$$= 0.5\left\{e^{\beta x} \operatorname{erfc}\left(\frac{x}{\sqrt{4\tau}} + \beta\sqrt{\tau}\right) + e^{-\beta x} \operatorname{erfc}\left(\frac{x}{\sqrt{4\tau}} - \beta\sqrt{\tau}\right)\right\}$$

*Piezometric head, seepage, and bank storage formulas.* The fluctuation of the piezometric head, the rate of seepage $q$ per unit front of stream, and the corresponding bank storage $V$ during a period $t_0$ since the sudden change of water levels (the subsequent relations pertain to the water-table case if $b$ is replaced with $h_0$) are finally obtained as

$$h - h_0 = 2\delta \sum_{m=1}^{\infty} R_0 \cos\frac{b - z}{b}\beta_m\left\{\exp(\beta_m x/b) \operatorname{erfc}\left(\frac{x}{\sqrt{4\nu t}} + (\beta_m/b)\sqrt{\nu t}\right)\right.$$
$$\left. + \exp(-\beta_m x/b) \operatorname{erfc}\left(\frac{x}{\sqrt{4\nu t}} - (\beta_m/b)\sqrt{\nu t}\right)\right\} \quad (34)$$

$$q = -K\int_0^b \partial/\partial x[h(0, z, t) - h_0]\, dz$$
$$= \frac{4K\delta}{\sqrt{\nu t}} \sum_{m=1}^{\infty} R_0\cdot(b/\beta_m) \sin \beta_m\{(\beta_m/b)\sqrt{\nu t} + \operatorname{ierfc}[(\beta_m/b)\sqrt{\nu t}]\} \quad (35)$$

and

$$V = \int_0^{t_0} q\, dt = 2K\delta/\nu \sum_{m=1}^{\infty} R_0\cdot(b/\beta_m)^2 \sin\beta_m\{1 + 2\nu t_0(\beta_m/b)^2 - 4i^2 \operatorname{erfc}[(\beta_m/b)\sqrt{\nu t_0}]\}$$
$$(36)$$

in which

$$R_0 = (\sin\beta_m)/(2\beta_m + \sin 2\beta_m) \quad (37)$$

$\beta_m$ are the roots of (15), and ierf and $i^2$ erfc are, respectively, the first and second repeated integrals of the error function, tabular values for which are available [*Carslaw and Jaeger,* 1959].

*Comparison with approximate available solutions.* When $(b/B) \leq 0.10$, the relations (34) through (37) will reduce, after making use of the approximations stated in the paragraph preceding equation 25, to the following relations, respectively:

$$h - h_0 = \frac{\delta}{2}\left\{e^{x/B} \operatorname{erfc}\left(\frac{x}{\sqrt{4\nu t}} + \sqrt{\nu t/B^2}\right) + e^{-x/B} \operatorname{erfc}\left(\frac{x}{\sqrt{4\nu t}} + \sqrt{\nu t/B^2}\right)\right\} \qquad (38)$$

$$q = Kb\ \delta/\sqrt{\nu t}\left\{\sqrt{\nu t/B^2} + \operatorname{ierfc}\left(\sqrt{\nu t/B^2}\right)\right\} \qquad (39)$$

and

$$V = \frac{Kb\ \delta B}{2\nu}\{1 + 2(\nu t_0/B^2) - 4i^2\operatorname{erfc}\left(\sqrt{\nu t_0/B^2}\right)\} \qquad (40)$$

which relations are identical to those obtained by *Hantush* [1962] by using the approximate theory, on observing that

$$\operatorname{ierfc}(x) = e^{-x^2}/\sqrt{\pi} - x \operatorname{erfc}(x)$$

$$4i^2 \operatorname{erfc}(x) = (1 + 2x^2)\operatorname{erfc}(x) - 2/\sqrt{\pi}\ xe^{-x^2}$$

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x)$$

and

$$B^2 = T/(K'/b')$$

## CONCLUDING REMARKS

Solutions for several problems in leaky aquifers are obtained by considering the effects of leakage on the flow as a boundary condition. The same procedure may be followed in obtaining solutions for other flow problems. This approach, preserving the vertical flow components, yields more general solutions than those obtained by using the approximate theory where the effect of leakage is incorporated in an approximate differential equation. However, the many solutions that have already been obtained by using the approximate theory yield results that are in very close agreement with the more general counterpart solutions wherever the thickness $b$ of the aquifer relative to the leakage factor $B$ of the flow system is less than 0.10, $B$ being equal to $\sqrt{T/K'/b'}$) Fortunately, this criterion is satisfied by many natural leaky systems. The Roswell basin in New Mexico is one of such systems. Accordingly, the simple solutions now in use are applicable in this basin as well as in many other basins. There are other instances where these simple formulas are not applicable, however. For example, in a leaky system having values of $T$ = 1510 gpd/ft and $K'/b'$ = 7.85 × $10^{-3}$, and hence, $B$ = 440 ft, these simple formulas are inapplicable where the aquifer thickness is greater than 44 feet.

*Acknowledgment.* This research is supported in part by funds provided by the U. S. Department of the Interior as authorized under Water Resources Research Act of 1964, Public Law 88-379.

## REFERENCES

Carslaw, H. S., and J. C. Jaeger, *Conduction of Heat in Solids,* Oxford University Press, London and New York, 2nd ed., 1959.

DeWiest, R. J. M., *Geohydrology,* John Wiley & Sons, New York, 1965.

Dwight, H. B., *Mathematical Tables of Elementary and Some Higher Mathematical Functions,* Dover Publications Inc., New York, 1961.

Hantush, M. S., Analysis of data from pumping tests in leaky aquifers, *Trans. Am. Geophys. Union, 37,* 702–714, 1956.

Hantush, M. S., Non-steady flow to a well partially penetrating a leaky aquifer, *Proc. Iraqi Scient. Soc., 1,* 10–19, 1957.

Hantush, M. S., Modification of the theory of leaky aquifers, *J. Geophys. Res., 65,* 3713–3725, 1960.

Hantush, M. S., Discussion of intercepting well in artesian aquifers, by Dean F. Peterson, *Trans. Am. Soc. Civil Eng., 127*(3), 39, 1962.

Hantush, M. S., Hydraulics of wells, in *Advances in Hydroscience,* edited by V. T. Chow, Academic Press, New York, 1964.

Hantush, M. S., Flow to wells in aquifers separated by a semi-pervious layer, *J. Geophys. Res.,* in press, 1967.

Hantush, M. S., and C. E. Jacob, Non-steady radial flow in an infinite leaky aquifer, *Trans. Am. Geophys. Union, 36,* 95–100, 1955.

Jacob, C. E., Radial flow in a leaky artesian aquifer, *Trans. Am. Geophys. Union, 27,* 198–205, 1946.

Schoeller, H., *Les Eaux Souterraines,* Masson et Cie., Paris, 1962.

(Manuscript received November 14, 1966; revised December 16, 1966.)